

Requirement Analysis

Technology Stack (Architecture & Stack)

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below along with the information in Tables 1 and 2.

Visualizing Housing Market Trends – Architecture Flow

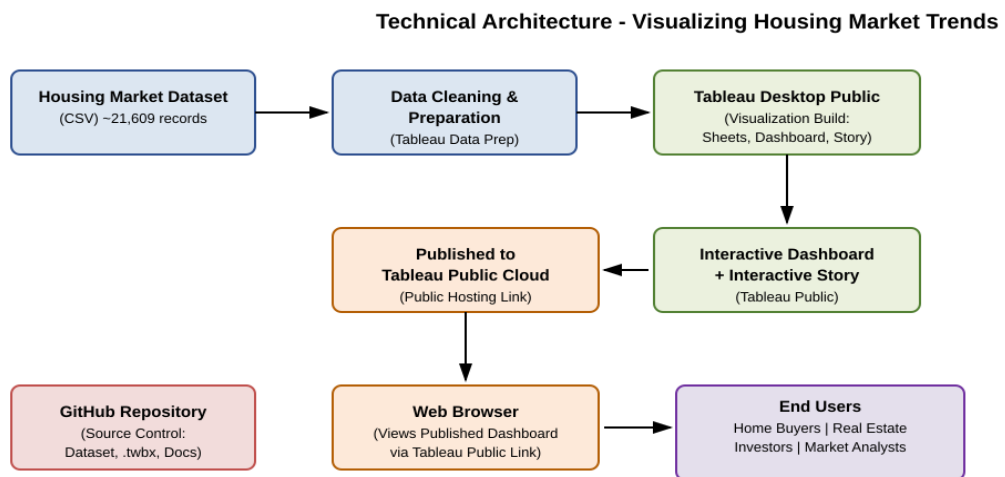


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How stakeholders view the finished dashboard and story through a web browser	Tableau Public (Web View), Web Browser
2.	Application Logic-1	Cleans and prepares the raw housing CSV dataset prior to visualization (handling missing values, converting fields, binning)	Tableau Data Prep / Excel
3.	Application Logic-2	Builds the individual visualizations, the 4-tile dashboard, and the storyboard from the prepared dataset	Tableau Desktop Public
4.	Application Logic-3	Publishes the finished dashboard and story to Tableau Public so it can be viewed via a shareable web link	Tableau Public (Publish/Share)

5.	Database	This project does not use a traditional database; data is sourced directly from a CSV file	Not Applicable for this project
6.	Cloud Database	Not Applicable for this project	Not Applicable for this project
7.	File Storage	Storage of the raw and cleaned housing dataset and the Tableau workbook	Local File System (CSV, .twbx), Google Drive (shared team storage)
8.	External API-1	Not Applicable for this project	Not Applicable for this project
9.	External API-2	Not Applicable for this project	Not Applicable for this project
10.	Machine Learning Model	Not Applicable for this project	Not Applicable for this project
11.	Infrastructure (Server / Cloud)	Development done locally in Tableau Desktop; finished dashboard hosted on Tableau Public (cloud). Source code and documentation version-controlled via Git and hosted on GitHub	Local Machine, Tableau Public (Dashboard Hosting), GitHub

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Tools Used	Tools and platforms used to clean data, build visualizations, and host the final dashboard	Tableau Desktop Public, Tableau Public, GitHub
2.	Security Implementations	Public Tableau dashboard with controlled repository access through GitHub; no sensitive or personal data is exposed in the published dashboard	GitHub repository access control
3.	Scalable Architecture	Additional sheets, filters, or visualizations can be added to the workbook without changing the overall dashboard structure	Modular Tableau workbook (Sheets + Dashboard + Story)
4.	Availability	Dashboard and story are accessible anytime through the Tableau Public link, without dependency on a dedicated hosting server	Tableau Public hosting

5.	Performance	Optimized visualizations through Top-N filters, binned measures, and pre-aggregated views for faster rendering	Tableau Public rendering optimization
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